

pilha.cpp

```
1 //Nome: Guilherme Borges de Pádua Barbosa
2 //Nº USP: 15653045
3
4 #include <iostream>
5 #include "pilha.h"
6 using namespace std;
7
8 Stack::Stack(){
9     top = NULL;
10    count = 0;
11 }
12
13 Stack::~Stack(){
14     clear();
15 }
16
17 bool Stack::empty(){
18     return (top == NULL);
19 }
20
21 void Stack::isEmpty(){
22     if(empty()){
23         cout << "Pilha Vazia" << endl;
24         abort();
25     }
26 }
27
28 bool Stack::full(){
29     return false;
30 }
31
32 void Stack::push(StackEntry x){
33     StackPointer p;
34     count++;
35
36     p = new StackNode;
37     if(p == NULL){
38         cout << "Memória Insuficiente" << endl;
39         abort();
40     }
41
42     p->entry = x;
43     p->nextNode = top;
44     top = p;
45 }
46
47 void Stack::pop(StackEntry &x){
48     isEmpty();
49     x = top->entry;
50
51     StackPointer p;
```

```
52     p = top;
53     top = top->nextNode;
54     delete p;
55 }
56
57 void Stack::clear(){
58     StackEntry temp;
59
60     while(!empty()){
61         pop(temp);
62     }
63
64     count = 0;
65 }
66
67 int Stack::size(){
68     return count;
69 }
70
71 void Stack::getTop(StackEntry &x){
72     isEmpty();
73
74     pop(x);
75     push(x);
76 }
77
```